

AltSports Surf Location Analysis Report

AUTHOR

Owen Loughery

Project Overview

Professional surfing is one of the most location-dependent sports in the world. Every event is held in a different wave environment, and those environments vary in ways that directly affect performance. Some locations reward long, flowing rail surfing; others reward commitment in heavy barrels; some are more predictable, while others are highly variable.

The goal of this project was to answer two related questions:

1. **Is there a relationship between event location and how well a surfer performs?**
2. **If that relationship exists, what helps explain it?**

My analysis suggests that event location does matter, but to figure out exactly why a location matters I found that the relationship appears to be driven largely by three deeper factors:

- whether an event resembles a surfer's **home environment**
- the **type of break** being surfed
- A surfers **past performance** at the location

To study this, I worked with historical WSL results, enriched the data with surfer nationality, stance, and break type. Then I made my interpretations of the effects by looking at Bet 365's surfers betting odds for the 2025 WSL tour event, as it is believed that they account for specific break types in their model and algorithm that calculates their betting odds. So the way I found whether there appears to be a relationship between the factors was not only between the variable and a surfer's score, but also looking into how it fits with these betting odds. I then compared performance across locations using a standardized score within each event.

Data and Feature Engineering

My report uses completed heats from WSL events, focusing primarily on Championship Tour Men's results for the main analysis. I standardized each surfer's heat score within an event so that performances could be compared across contests with different scoring environments.

A standardized score (`score_std`) was calculated as a z-score within each event:

$$score_{std} = \frac{score - \bar{x}_{event}}{sd_{event}}$$

This makes the score easier to compare across different events with: positive values indicating a surfer performed above the event average, while negative values indicate below-average performance.

I also classified each event location into one of four broad wave categories, creating a feature called (`breakType`):

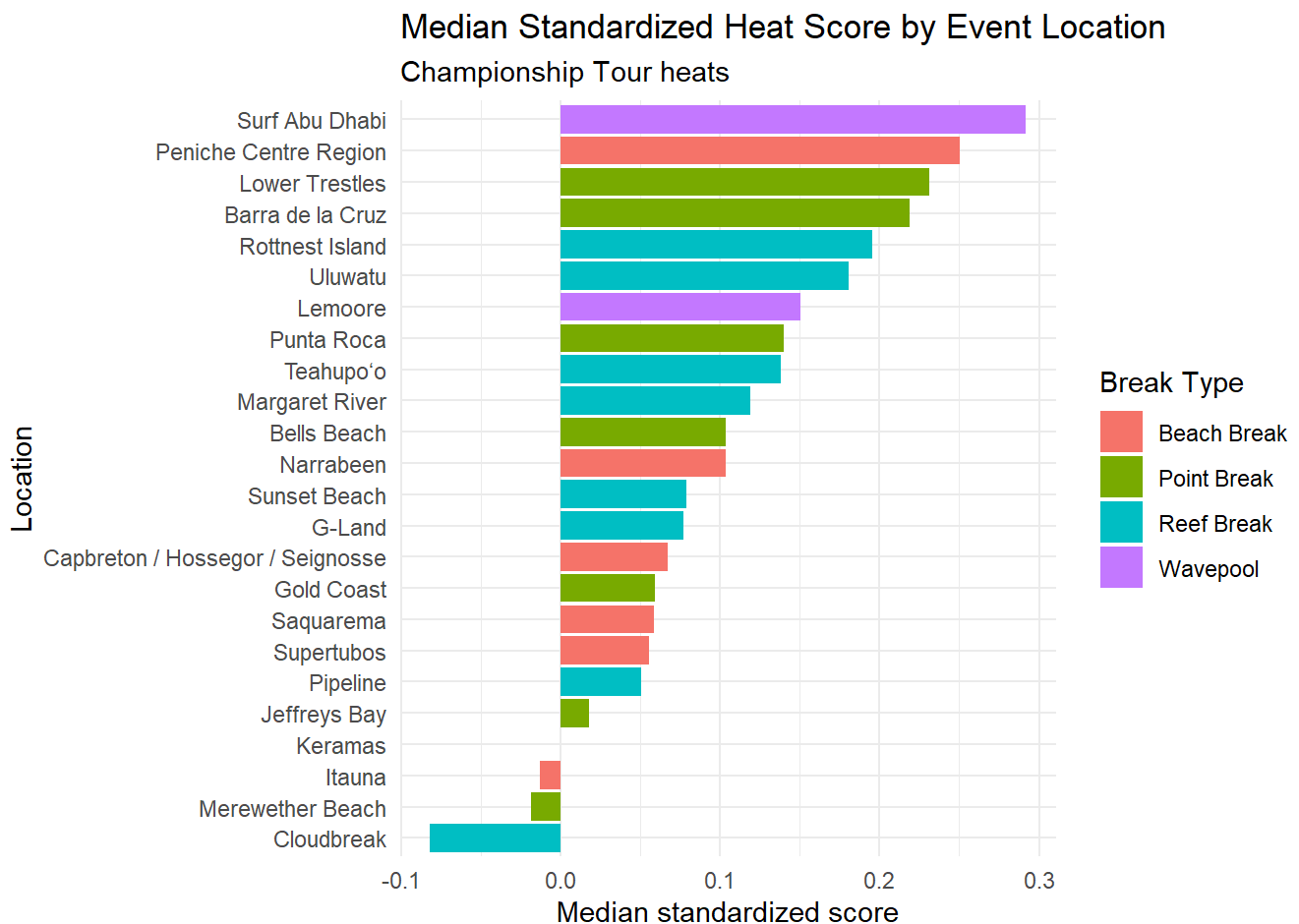
- Point Break
- Reef Break
- Beach Break
- Wave Pool

Finally, I created a **home break indicator** by matching a surfer’s nationality to the nation associated with each event location.

Research Question 1: Does Event Location Relate to Performance?

The first step was to look for evidence that location matters at all. If location did not have an effect on the performance, then surfers would tend to post similar standardized score distributions no matter where they competed. Instead, the data showed that many surfers have noticeably different score distributions across locations.

A useful way to visualize this is to compare median standardized performance by location across the tour.



This plot is not meant to rank venues in a definitive sense, but instead, it shows that event environments are not interchangeable. Performance distributions shift across locations, which supports the idea that **where an event is held is related to how surfers perform.**

Research Question 2: Why Does the Location Effect Exist?

Now that a location-performance relationship was visible, the next step is explaining it. The strongest explanations in my analysis were:

1. **Home-break familiarity**
2. **Break type**
3. **A surfer's past performance at the location**

Home-Break Effect

One possible reason location matters is that some surfers compete in environments closer to what they grew up surfing. A Brazilian at Saquarema, a Hawaiian at Pipeline, or an Australian at Bells Beach may benefit from familiarity with wave shape, pacing, section timing, and lineup behavior.

To explore that idea, I created a binary indicator for whether the event was effectively a home-nation break for the surfer.

This comparison helps explain why location matters: **some of the observed location effect seems to come from whether a venue aligns with a surfer's home surfing environment.**

Break Type Effect

The second explanation is that not all locations test the same kind of surfing. Point breaks, reef breaks, beach breaks, and wave pools reward different strengths. A surfer who excels in heavy reef barrels may not necessarily produce the same results at long, open point breaks, and vice versa.

This then supports the broader interpretation: **location matters partially because different locations are testing different surfing skill sets.** While this has an effect for some surfers, not all surfers are going to be as effected because they might just either be a lower end competitor that doesn't perform significantly better at different break types, or they are a consistent top competitor that does well regardless of the break type. So this is why the effect of the type of break appears to be significant for some surfers compared to not being as important for others in terms of how well they perform at a location.

Surfer's Past Performance

The third explanation was looking into how a surfer has done in a certain location in the past. This is important because surfers that consistently compete well at a certain location have a higher chance to compete well there again.

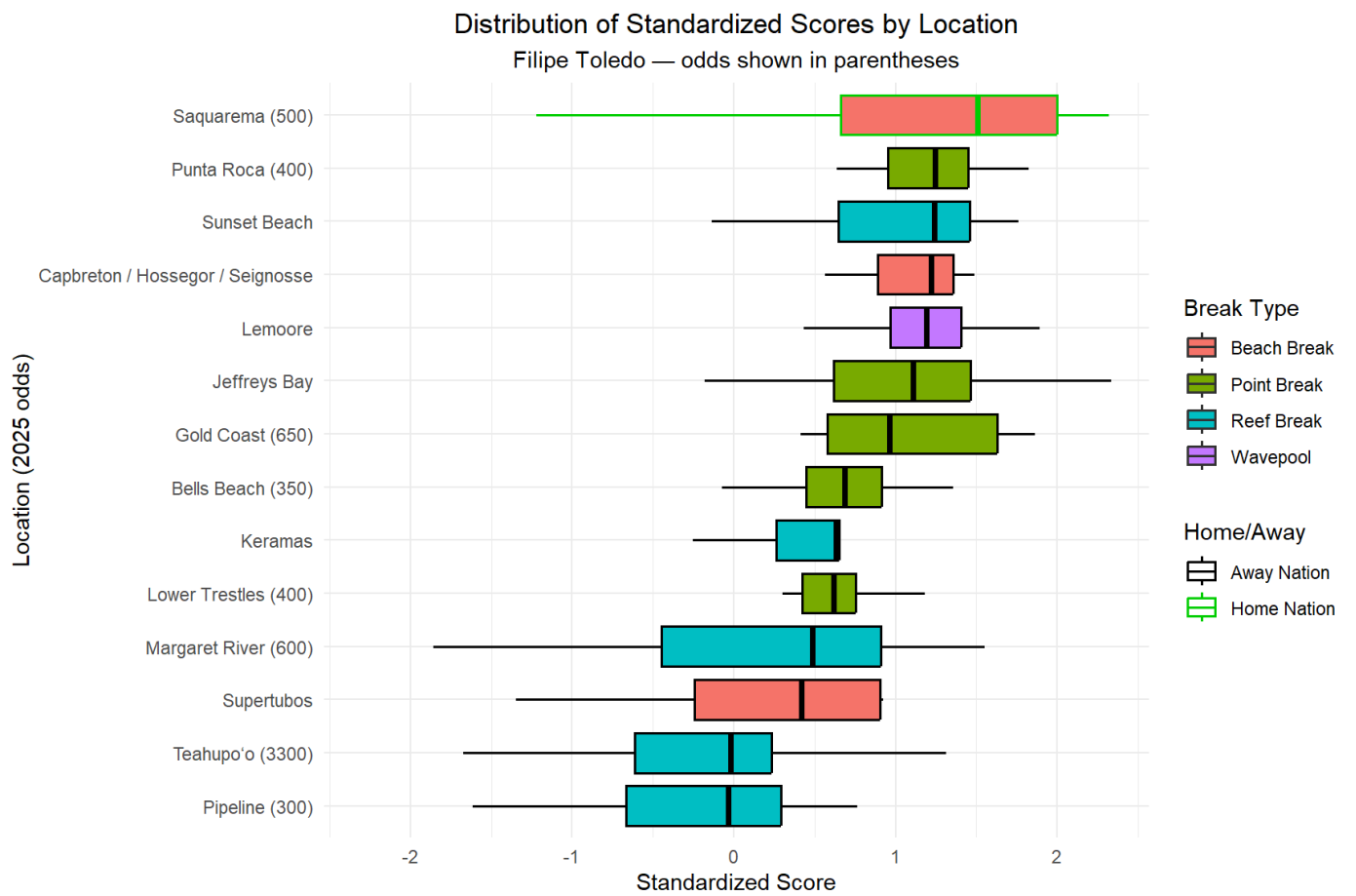
This seemed to be one of the most important factors in figuring out why location matters: as a surfer's past performance at an event has a big effect on how they are going to do the next time they compete there.

Evidence of these effects

Home-Break Effect

The following graphs and tests show evidence that different surfers compete better depending on the type of break. I also showed the betting odds and home/away in the graphs to visualize these as well but these effects were much more clear grouping by different locations instead of surfers like I have here which I will explain later. But, for the type of break, it is clear that there is a grouping happening for certain surfers with them having better scores for certain types of breaks compared to others.

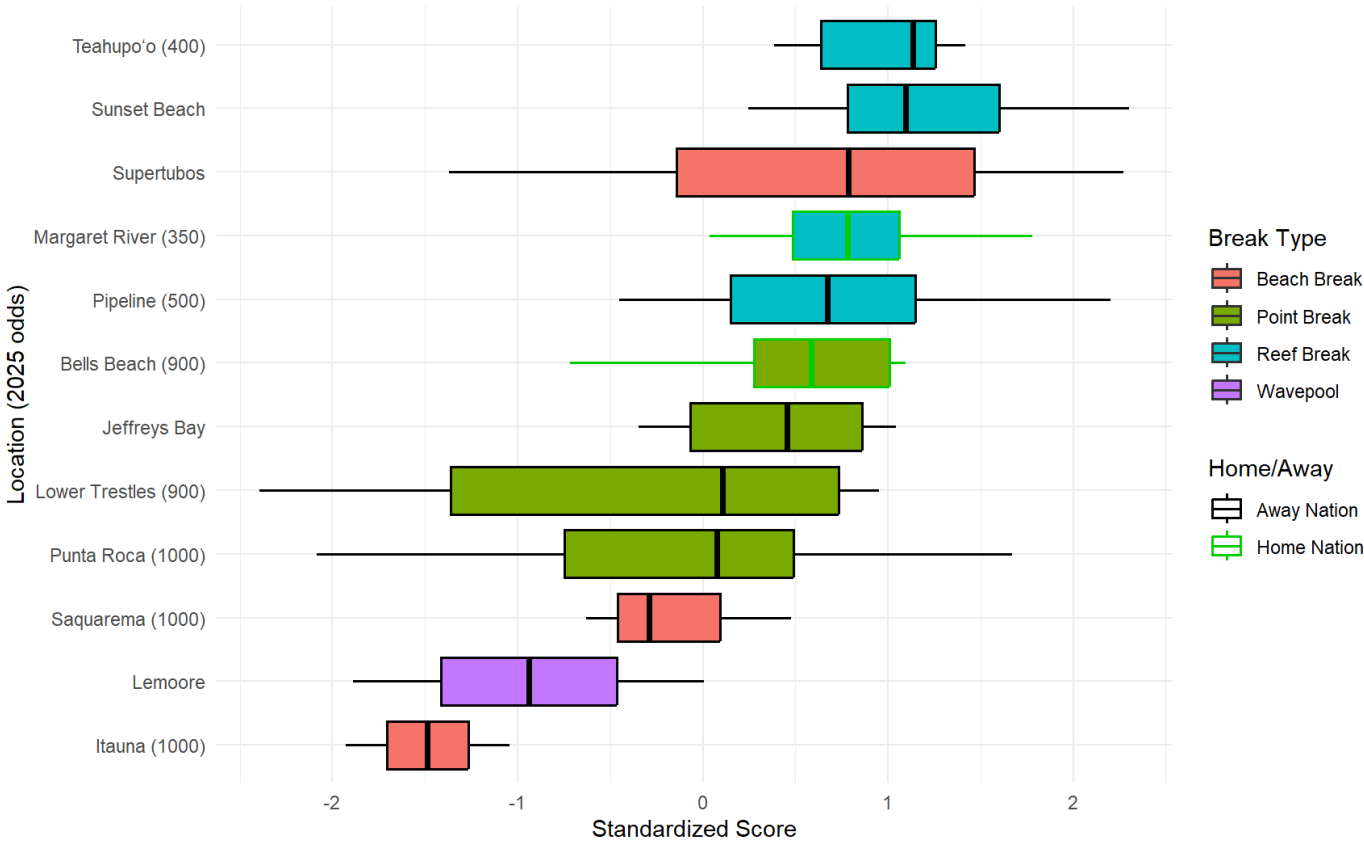
Filipe Toledo



Jack Robinson

Distribution of Standardized Scores by Location

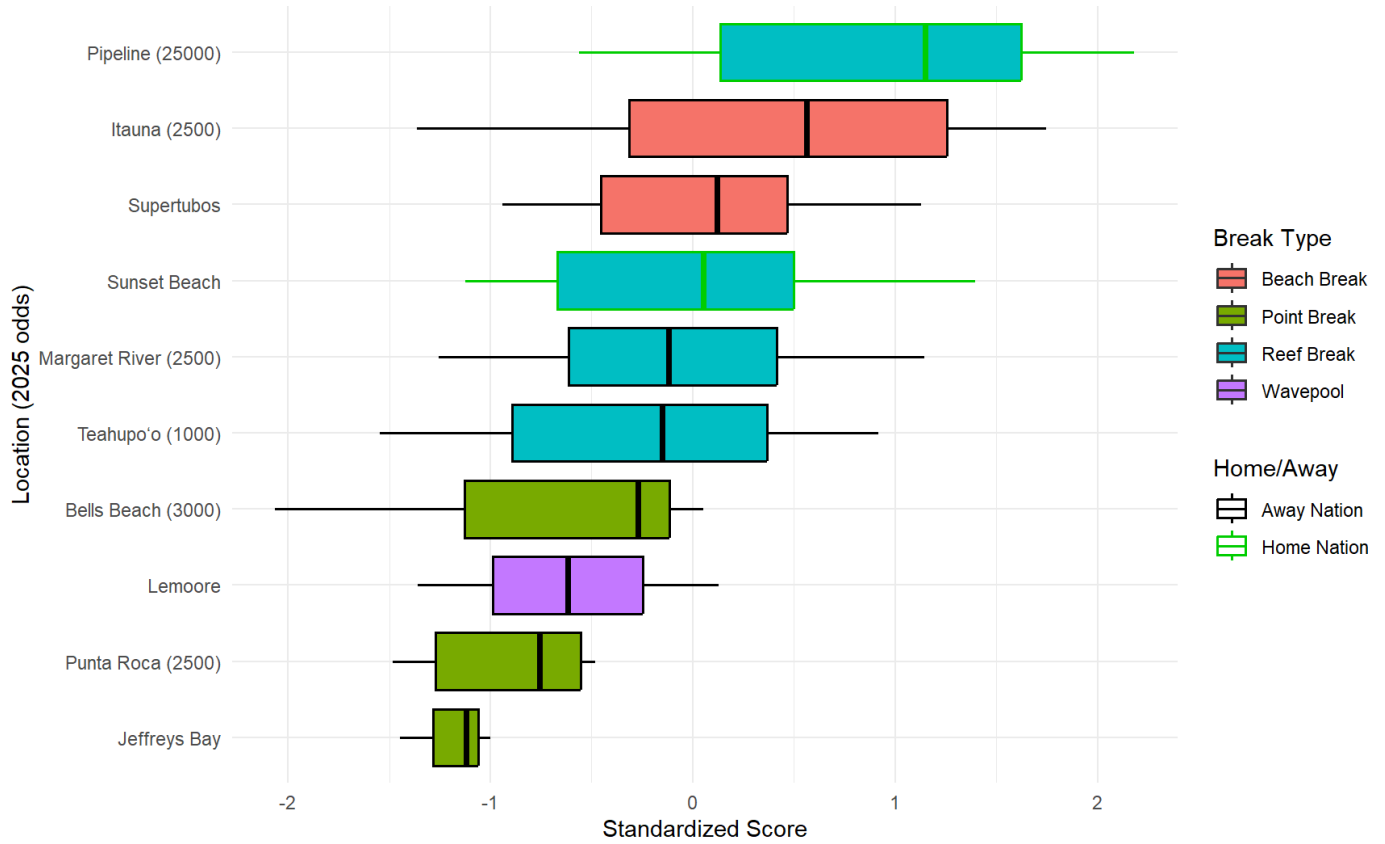
Jack Robinson — odds shown in parentheses



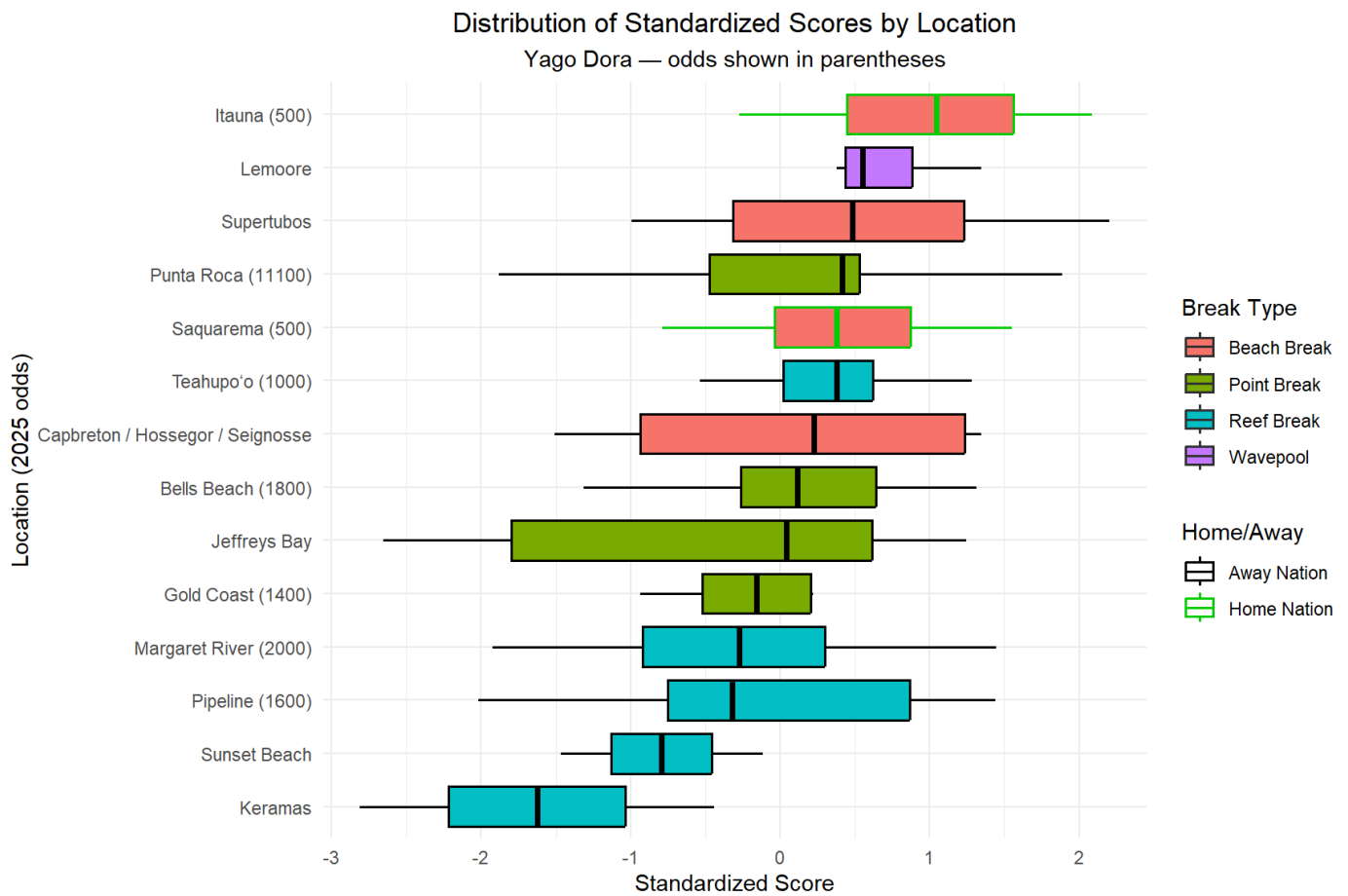
Barron Mamiya

Distribution of Standardized Scores by Location

Barron Mamiya — odds shown in parentheses



Yago Dora

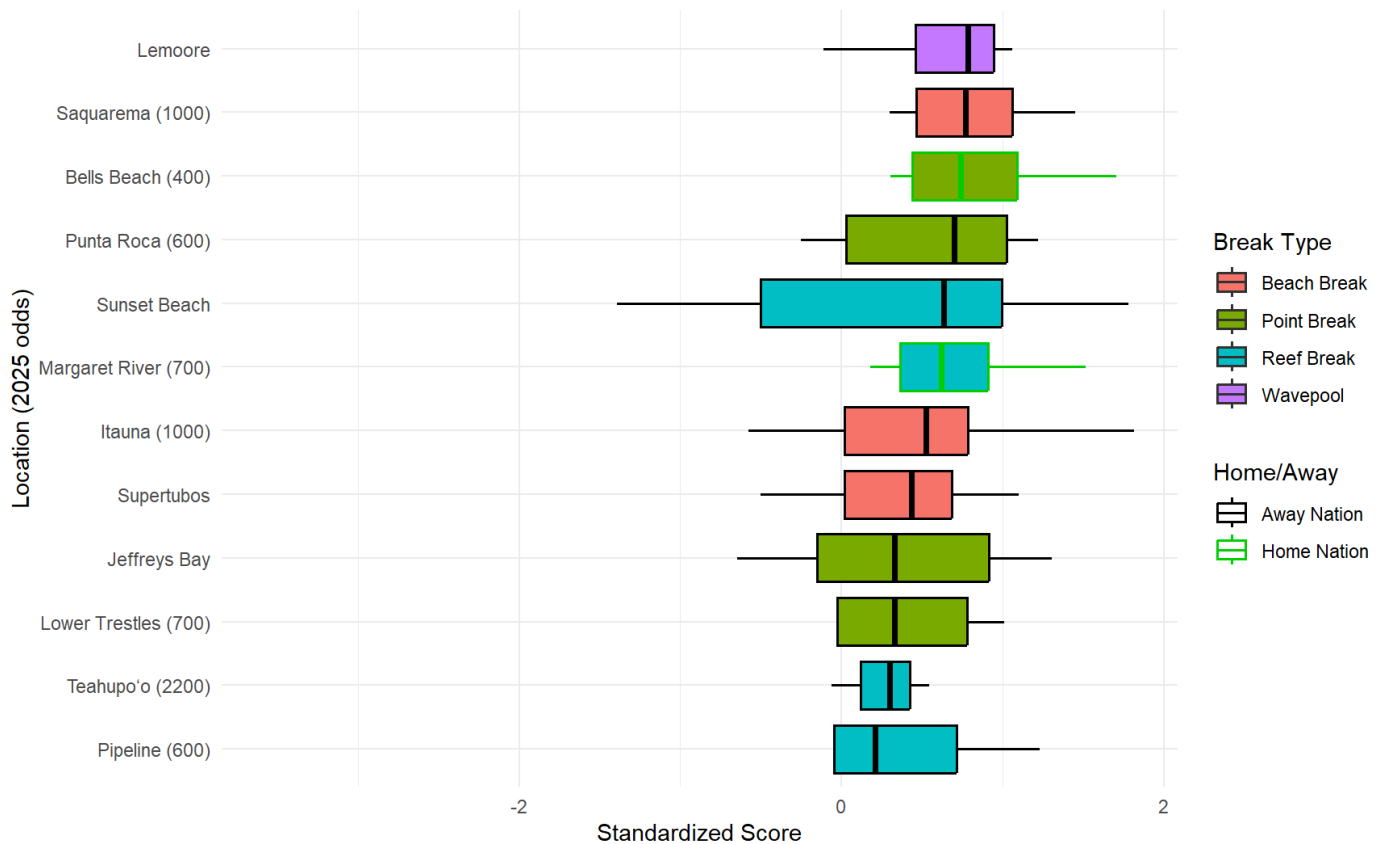


To give a quick example of a surfer that wouldn't have their score be effected, Ethan Ewing has won or finished in the top 5 of the tour overall multiple times meaning he is one of the best competitors in the more recent years. So to be able to consistently be a top level competitor, he has done well at all locations in the past which can be seen as his mean standardized score is positive for all locations. The effect of break type does not appear to have much of an effect on his performance as since he overall does well regardless of the event location. So, this is an example of why break type wouldn't have an effect on all individual surfers.

Ethan Ewing

Distribution of Standardized Scores by Location

Ethan Ewing — odds shown in parentheses



To actually look into the significance of this effect, I decided to do an ANOVA test on the effect of the break type on the standardized score, **but accounting for individual surfers** so I did a two way ANOVA with an interaction term. This is because the relationship is very dependent on the individual surfer as I explained earlier, so looking at this relationship for each individual surfer instead of overall shows this effect.

Two-factor ANOVA for standardized score with surfer, break type, and their interaction

Term	Df	Sum Sq	Mean Sq	F value	Pr(>F)
athlete_name	145	629.5715538	4.3418728	5.0578085	0.0000
breakType	3	0.9529051	0.3176350	0.3700102	0.7750
athlete_name:breakType	237	261.4357292	1.1031043	1.2849963	0.0027
Residuals	4076	3499.0398118	0.8584494	NA	NA

From this ANOVA I looked into if there is an interaction effect between whether the effect of the break type on a surfer's score depends on who the surfer is. Since we get a significant surfer-by-break-type interaction p-value of .0027 which is $< .05$, this indicates that break type matters differently for different surfers. In other words, performance depends less on break type in general and more on the match between a specific surfer and a specific break environment which supports my earlier claims.

To further demonstrate this I found which surfers were individually effected by the type of break. To do this I found which ones had significant p-values from a one way ANOVA that compares the mean standardized score at each break type level for an individual surfer. Here are the results:

Surfers whose performance appears especially sensitive to break type

athlete_name	breakType_pval
Barron Mamiya	0.0002042
Filipe Toledo	0.0011521
Alejo Muniz	0.0050532
Wiggolly Dantas	0.0057179
Michel Bourez	0.0062737
Kauli Vaast	0.0062764
Jack Robinson	0.0223299
Jesse Mendes	0.0232335
Yago Dora	0.0269850
Kanoa Igarashi	0.0465470
Caio Ibelli	0.0827513
Griffin Colapinto	0.0908738
Jeremy Flores	0.0980964

This then proves my claims about how certain surfers are much more effected by the break type than others as only these surfers had significant p-values (at an alpha level of .1), so to visualize this here are graphs I described before for a few of these surfers showing how the surfer appears to perform better at certain break types compared to others:

Break Type Effect + Surfer's Past Performance

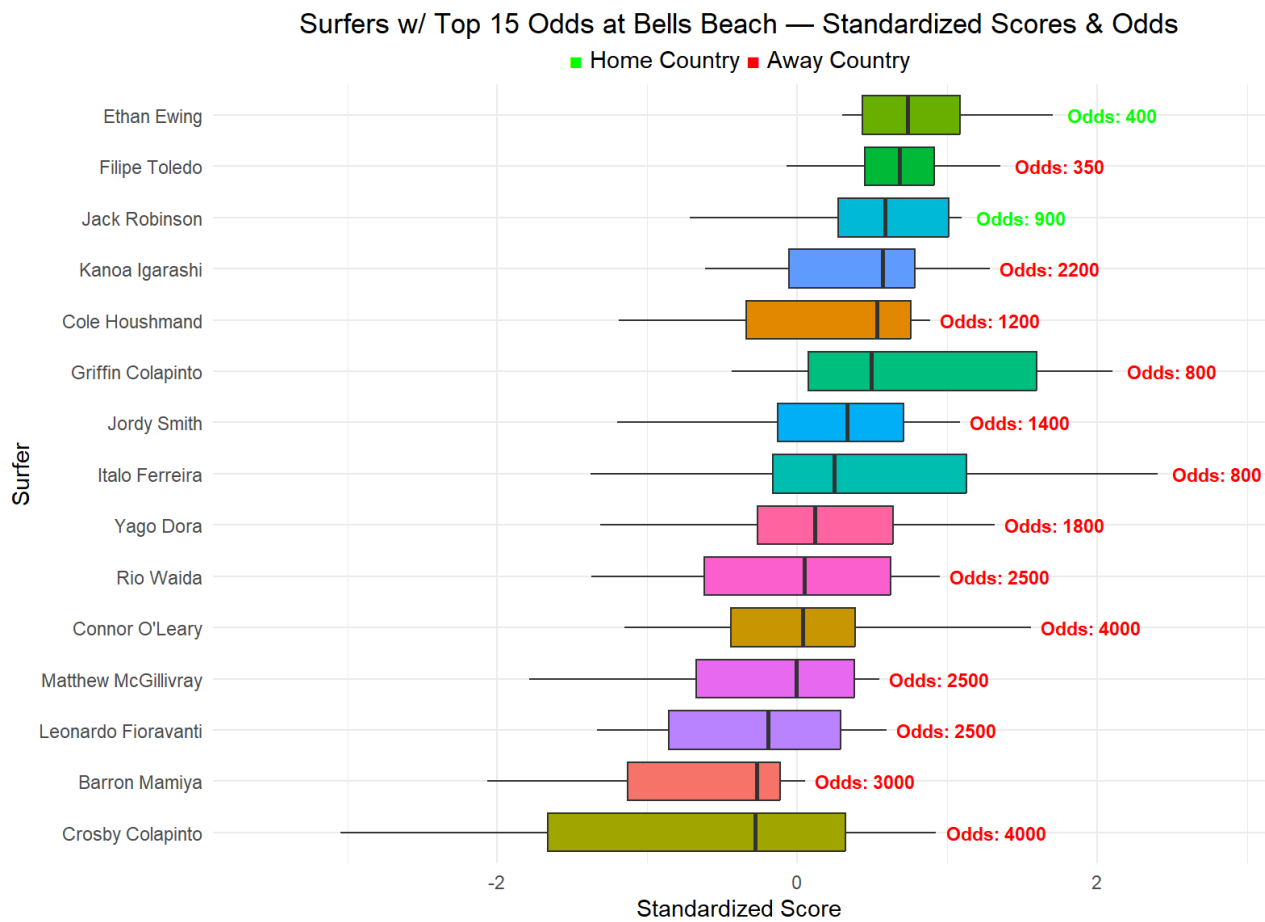
The next effects I looked into together are the **home break effect** and how a surfer performed at a location **in the past**. To look into this effect I decided to look into visualizations of the different event locations and display — 1. whether it is a home break or not 2. the odds from the Bet365's 2025 year betting odds — for box plots of the top 15 surfers standardized scores for each location. The reason I did this was because this shows **how well a surfer has competed in comparison to other surfers in the past for each location**. The results then clearly show two important things.

1. How the Bet365's betting odds are much better (lower numbers) for surfers that have competed well in the past at each location. Showing how it appears that a big part of the Bet365's betting odds model and algorithm utilize a feature examining how a surfer has done at a location in the past to determine their betting odds. Another thing to note is that surfer's that have competed well in the past at this location but still have poor odds (high numbers) typically tended to be surfer's that are overall ranked lower on the tour. So, even though they have worse odds in comparison to other surfers that are around the same standardized score for that location, their odds are much greater than if they were competing anywhere else where they typically don't do well. (i.e. George Pittar who typically does not do well on the tour but is really good at Margaret river, has betting odds of 3300 but typically has betting odds of 10,000+)
2. It appears that surfers that perform the best at certain locations tend to have the location be a break from the nation they are from. It was a little hard to show this through the visualizations entirely as there is a lot of missing

event information for some surfers at different locations, so I gave summaries of the amount of events they have competed in at each location and the number of heats competed to show weird values that might just be explained by missing data. (Note: it is much more important to have a high number of events competed than heats as this is similar to the number of repetitions completed).

Here are some example locations to show these findings:

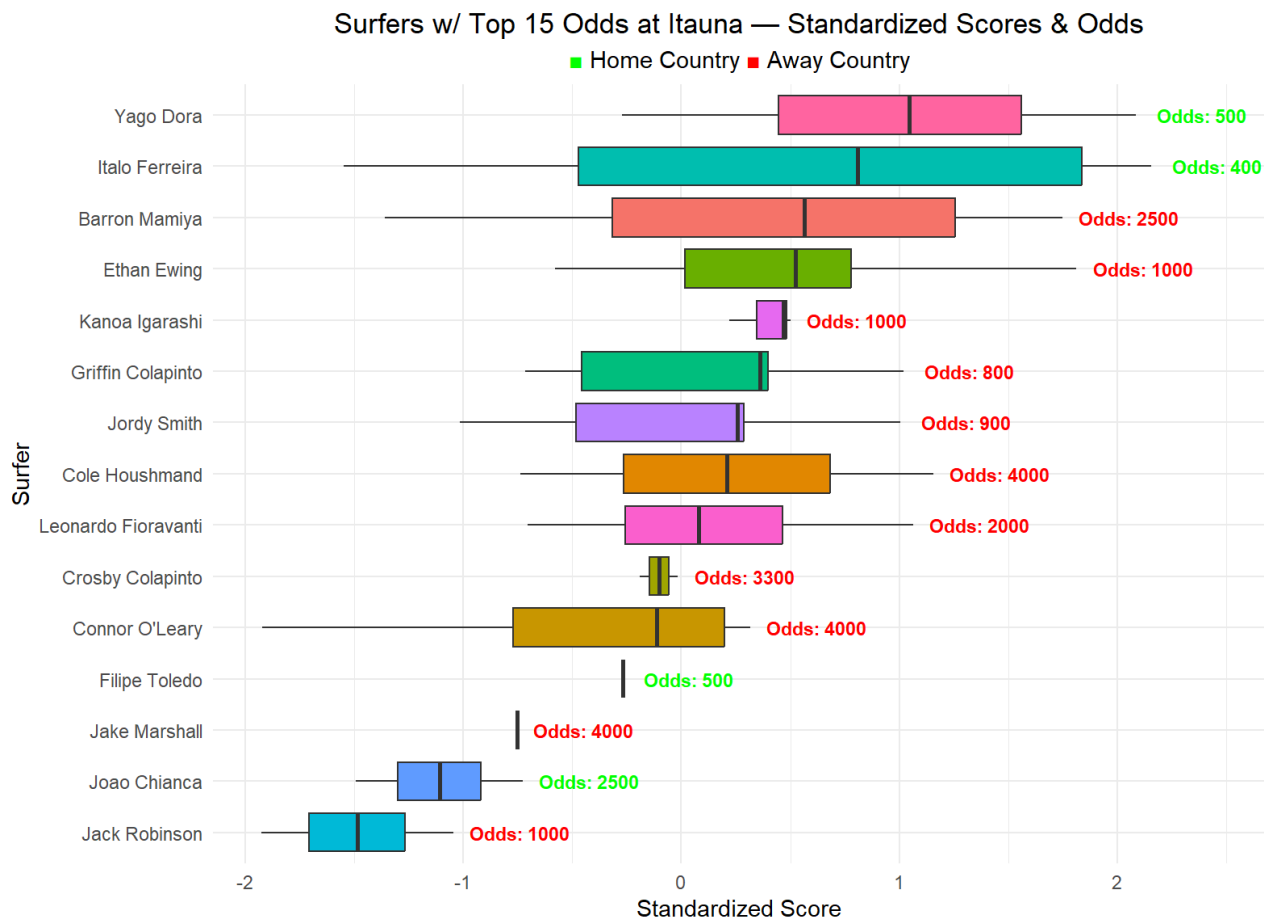
Bells Beach



Surfer Name	# of Events Completed	# of Heats Completed
Italo Ferreira	6	15
Kanoa Igarashi	6	13
Jordy Smith	6	12
Filipe Toledo	5	18
Griffin Colapinto	5	15
Connor O’Leary	5	11
Yago Dora	5	8
Leonardo Fioravanti	5	6
Ethan Ewing	4	15
Jack Robinson	4	11

Surfer Name	# of Events Completed	# of Heats Completed
Matthew McGillivray	4	11
Barron Mamiya	4	6
Rio Waida	3	7
Cole Houshmand	2	7
Crosby Colapinto	2	3

Itauna

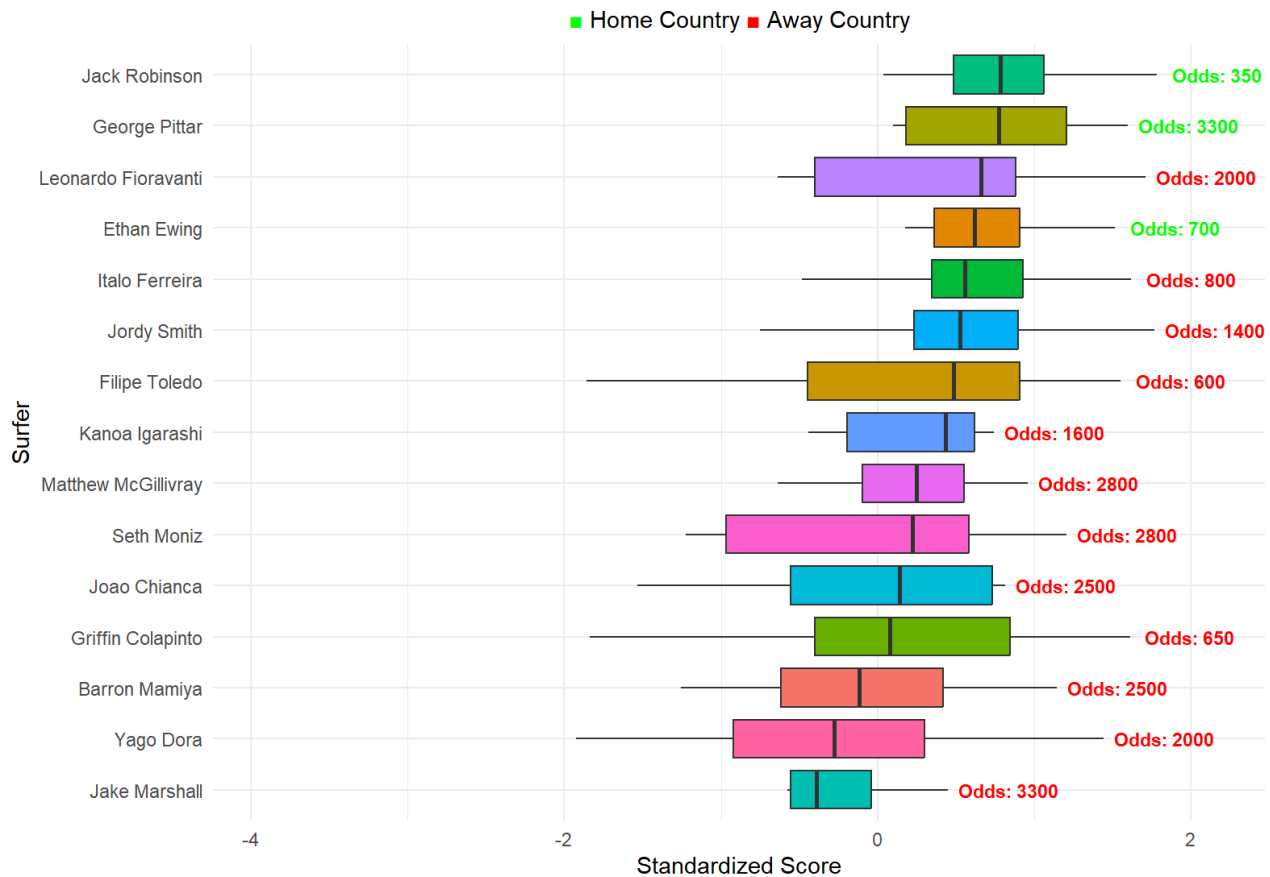


Surfer Name	# of Events Completed	# of Heats Completed
Yago Dora	2	8
Ethan Ewing	2	6
Griffin Colapinto	2	5
Italo Ferreira	2	5
Jordy Smith	2	5
Barron Mamiya	2	4
Connor O'Leary	2	4
Leonardo Fioravanti	2	4

Surfer Name	# of Events Completed	# of Heats Completed
Kanoa Igarashi	2	3
Jack Robinson	2	2
Joao Chianca	2	2
Cole Houshmand	1	2
Crosby Colapinto	1	2
Filipe Toledo	1	1
Jake Marshall	1	1

Margaret River

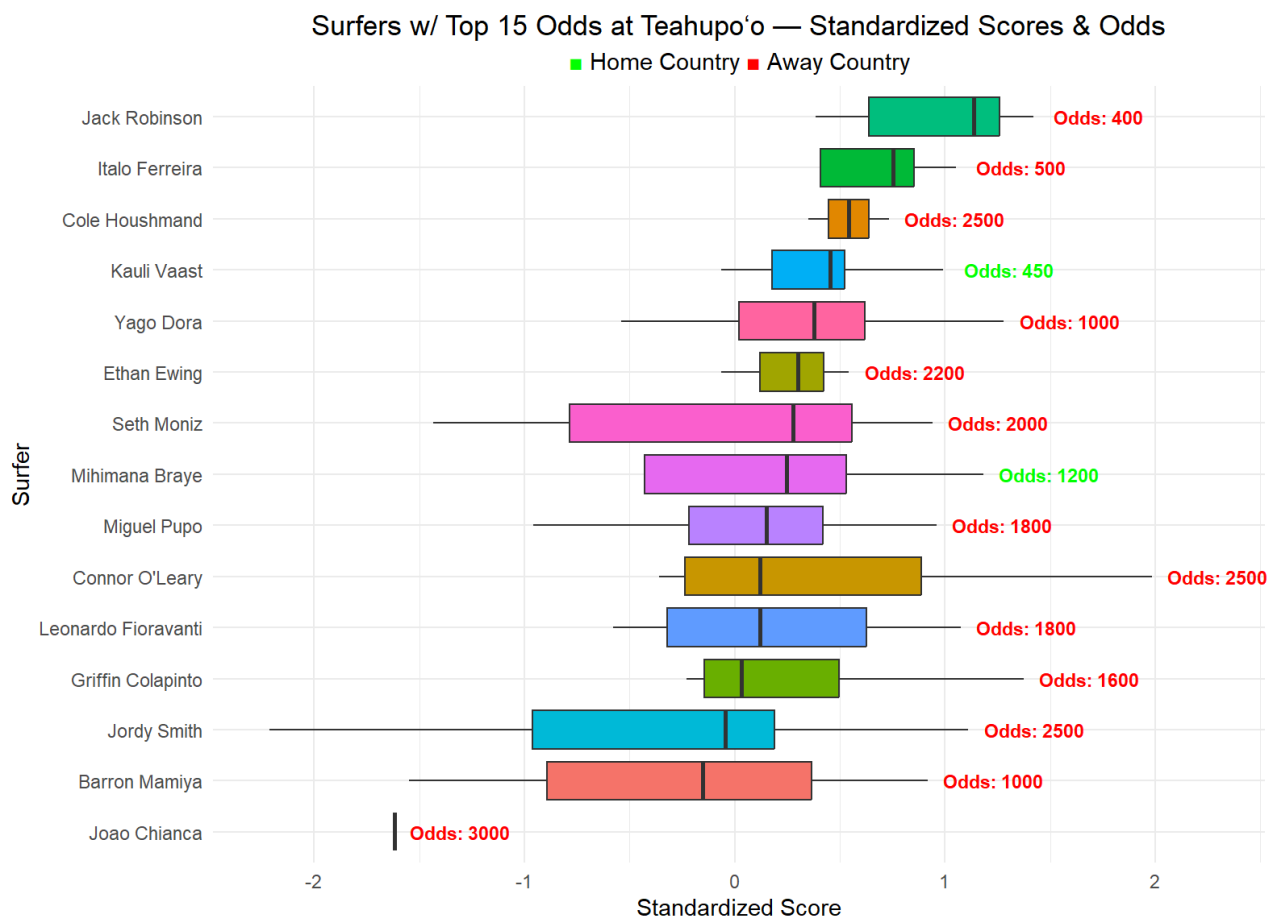
Surfers w/ Top 15 Odds at Margaret River — Standardized Scores & Odds



Surfer Name	# of Events Completed	# of Heats Completed
Griffin Colapinto	6	21
Jordy Smith	6	21
Italo Ferreira	6	14
Seth Moniz	6	13
Leonardo Fioravanti	6	11
Kanoa Igarashi	6	10

Surfer Name	# of Events Completed	# of Heats Completed
Jack Robinson	5	16
Matthew McGillivray	5	14
Filipe Toledo	5	12
Ethan Ewing	5	11
Yago Dora	5	8
Barron Mamiya	4	11
Jake Marshall	4	6
Joao Chianca	3	7
George Pittar	2	5

Teahupo‘o



Surfer Name	# of Events Completed	# of Heats Completed
Yago Dora	5	11
Jordy Smith	5	10
Griffin Colapinto	5	8
Italo Ferreira	4	9

Surfer Name	# of Events Completed	# of Heats Completed
Seth Moniz	4	8
Connor O’Leary	4	5
Kauli Vaast	3	9
Barron Mamiya	3	7
Jack Robinson	3	6
Miguel Pupo	2	6
Leonardo Fioravanti	2	4
Mihimana Braye	2	4
Ethan Ewing	2	3
Cole Houshmand	1	2
Joao Chianca	1	1

Next I decided to further prove the effect of a surfers home break on how well they perform. Due to the lack of event history for exploring this type of relationship, I decided to restrict the analysis for this to surfer-location pairs with at least three completed events at each location. To test whether competing at a home break was associated with improved performance, I fit a two-way ANOVA using standardized score as the response and home-break status, location, and their interaction as predictors.

Two-way ANOVA for standardized score using home-break status, location, and their interaction

Term	Df	Sum Sq	Mean Sq	F value	Pr(>F)
home_break	1	7.0025	7.0025	7.2639	0.0071
locationName	11	17.1744	1.5613	1.6196	0.0869
home_break:locationName	11	17.1426	1.5584	1.6166	0.0877
Residuals	2133	2056.2492	0.9640	NA	NA

Standardized score summary for home vs. away breaks after restricting to surfer-location pairs with at least 3 events

home_break	mean_score_std	median_score_std	n
Away break	0.1196	0.2326	1776
Home break	0.2690	0.3392	381

From the ANOVA results, the main effect of home-break status was statistically significant at the .05 level ($p = 0.0071$), indicating that performance differed between home and away breaks.

Combined with the summary statistics, this result suggests that surfers performed better on average at home breaks than at away breaks. This is because the mean standardized score was 0.2690 for home breaks compared to 0.1196 for away breaks, and the median standardized score was also higher at home breaks (0.3392 vs. 0.2326). This suggests that competing at a home break is associated with stronger relative performance.

The interaction between home-break status and location was not statistically significant at the .05 level ($p = 0.0877$), although it was somewhat suggestive. Overall, these results provide the strongest evidence for a general home-break advantage once the sparse surfer-location histories are excluded, as when I did tests before, these sparse results caused there to be no significant results found. But, that was misleading as there was not enough information for certain surfers-location histories to be included which is why I made this separation.

Overall, the **home-break familiarity**, **break type**, and a surfer's **past performance** at the location appeared to be the biggest factors in explaining why surfers perform better at different locations.

What I Chose Not to Emphasize

I also explored surfer stance as a possible explanation for the location effect. While there were isolated locations where stance seemed relevant, it did not appear to explain the broader pattern nearly as well as the other explanations. Because of that, I decided that stance did not make up an important factor to be included.

Conclusion

This project began with a simple question: **does event location correlate with surfer performance?** The answer from this analysis appears to be **yes**.

But the more important result is *why* that relationship exists. The location effect appears to reflect deeper structural differences in the surfing environments being contested on how a surfer performs.

Three explanations stood out most clearly:

- surfers often perform better when competing in environments that are their **home breaks** (location in their home nation)
- surfers results vary across **break types**, suggesting that different venues reward different competitive strengths
- surfers **past performance** at a location has a big effect on how well they are predicted to do the next time they compete there, especially supporting how Bet365's betting odds better capture this difference

So, while "location" is the visible variable, the more informative interpretation is that event performance is tied to a combination of these features.